

Paper U-FourPostulates

Four-QM-Postulates Unification under Participation Measure

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of the four QM postulates from nothing. Result is conditional on the 13 ED primitives (Paper_087) + declared paper-specific postulates across constituent Phase-1 theorems.
2. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton, Wolfram Ruliad, causal-set program), not the operational-reconstruction lineage.
3. It does **not** override Paper_095's three-tier verdict grammar. Verdict for this paper is stated in §1.
4. It does **NOT** claim closed-proof reconstruction of the four QM postulates — this paper explicitly disavows the Hardy 2001 / CDP 2011 / Coecke-Kissinger framing. The synthesis is **compositional**, not reductive.
5. Phase-1 supplies **compositional synthesis at M2** with declared per-theorem postulates spanning the constituent papers (Papers_002, 005, 006, 007, U1–U5). Per-theorem substrate-derivation closure is OPEN. Numerical content ($\hbar$, specific Hamiltonians, observable spectra) INHERITED.

Abstract

This paper synthesizes the four standard QM postulates — (i) state-as-Hilbert-vector, (ii) observables-as-self-adjoint-operators, (iii) Born rule, (iv) unitary evolution + projective measurement — under the substrate-level participation measure $P_K = b_K e^{i\pi_K}$ (Paper_U1) and the V1/V5 kernel calculus. The synthesis is **compositional synthesis at M2, NOT closed-proof reconstruction** in the Hardy 2001 / CDP 2011 / Coecke-Kissinger sense. Distinct primitive routes underlie each postulate (P02 → state; Stone-on-P03/P13 → observables / evolution; P04-bandwidth → Born rule; P11 → collapse); per-theorem substrate-derivation closure is OPEN. Verdict: **M2** (compositional synthesis with declared per-theorem postulates) per Paper_095's three-tier verdict grammar. Key falsifier **F1**: refutation of any constituent Phase-1 derivation propagates upward to refute the synthesis.

1 Statement of Result

Claim. The four standard postulates of quantum mechanics — (i) state-as-Hilbert-vector, (ii) observables-as-self-adjoint-operators, (iii) Born rule for probabilities, (iv) projective measurement / unitary evolution — are **unified** under the substrate-level **participation measure** $P_K = b_K e^{i\pi_K}$ (Paper_U1) and the kernel calculus (V1 + V5).

Structural composition. Postulate I (state space) traces to P_K via P02; Postulate II (unitary evolution / observables) to Stone-on-P03/P13; Postulate III (Born rule) to participation-measure on P_K (Paper_002 / Paper_U1); Postulate IV (collapse) to P11. These are **distinct primitive routes**, not one source; the synthesis is **compositional**, not reductive.

The synthesis is at verdict **M2 (Intermediate Path C)** — substrate mechanisms identified across the four postulates; per-theorem constructive substrate-derivations are OPEN.

Verdict: **M2 (compositional synthesis with declared per-theorem postulates)**.

2 Primitive Inputs

2.1 Primitives invoked

The 13 primitives P01–P13 per Paper_087, all load-bearing in some constituent Phase-1 derivation.

2.2 Upstream Dependencies

- **I-002:** Born-Gleason rule.
- **I-005:** Projective measurement from P11 commitment.
- **I-006:** Unitary evolution from V1 kernel.
- **I-007:** Hilbert-space emergence as completion of motif algebra.
- **I-U1:** Participation measure $P_K = b_K e^{i\pi_K}$.
- **I-U2-Discrete + Continuum:** Sesquilinear inner-product.
- **I-U3:** Schrödinger equation via Stone.
- **I-U4:** Hamiltonian form.
- **I-U5:** Momentum operator.

3 The Four-Postulate Unification

3.1 Postulate I: State as Hilbert-space vector

Standard postulate: A pure quantum state is represented by a normalized vector $|\Psi\rangle$ in a complex separable Hilbert space $\mathcal{H}$.

Substrate-level derivation:

- Participation measure $P_K = b_K e^{i\pi_K} \in \mathbb{C}$ assigned to each motif K (Paper_U1).
- Sesquilinear inner-product (Paper_U2-Discrete + Paper_U2-Continuum) on the motif-space yields a Hilbert space.
- Hilbert-space emergence via completion of motif algebra (Paper_007).
- Normalization $\sum_K b_K^2 = 1$ (Born-Gleason, Paper_002).

The “state” is the participation measure on substrate motifs; the “Hilbert space” is its natural arena.

3.2 Postulate II: Observables as self-adjoint operators

Standard postulate: Physical observables are represented by self-adjoint operators on $\mathcal{H}$.

Substrate-level derivation:

- Position operator $\hat{x}$ (multiplicative; from P03 locus indexing).
- Momentum operator $\hat{p} = -i\hbar\nabla$ (translation generator; Paper_U5).
- Hamiltonian $\hat{H}$ (time-translation generator; Paper_U3).
- All are self-adjoint by Stone's theorem applied to substrate-level symmetry groups.

The “observables” are infinitesimal generators of substrate-level symmetry groups (P03 spatial, P13 temporal, etc.); self-adjointness is forced by Stone's theorem on the substrate Hilbert space.

3.3 Postulate III: Born rule

Standard postulate: Probability of outcome a on measuring observable $\hat{A}$ in state $|\Psi\rangle$ is $|\langle a|\Psi\rangle|^2$.

Substrate-level derivation:

- Born-Gleason rule (Paper_002): $\text{Prob}(K) = |P_K|^2 = b_K^2$.
- Substrate origin: P04 bandwidth content b_K^2 is the substrate-level participation density, which under operational measurement provides the empirical frequency = probability.

The Born rule is the **bandwidth-to-probability** identification at substrate level. It is form-COMPOSED by P04 + measurement-projection structure.

3.4 Postulate IV: Measurement / unitary evolution

Standard postulate (dual):

- **Unitary evolution** between measurements: $|\Psi(t)\rangle = U(t)|\Psi(0)\rangle$ with $U(t) = e^{-i\hat{H}t/\hbar}$.
- **Measurement-induced collapse:** measurement projects $|\Psi\rangle$ onto eigenspace of measured observable.

Substrate-level derivation:

- Unitary evolution: Paper_U3 (Stone's theorem) + Paper_006 (V1 kernel propagation).
- Measurement collapse: Paper_005 (P11 commitment-irreversibility producing the projection).

Both halves of Postulate IV emerge from substrate primitives: P11 (commitment) supplies the collapse direction; P13 (time homogeneity) supplies the unitary-evolution generator structure.

3.5 Why the four postulates are NOT independent in ED

In standard axiomatic QM, the four postulates are typically presented as **independent** axioms. In the ED substrate framework:

- **States and observables share a single substrate origin:** the participation measure P_K .
- **Born rule is the substrate-level identification** of $|P_K|^2$ with probability — not an independent axiom.
- **Unitary evolution and measurement-collapse share a single substrate origin:** P11 (commitment-irreversibility) supplies both the directional flow of unitary time-translation AND the irreversible collapse at measurement.

The four standard postulates are therefore **four facets of one substrate-level structure**. The unification is the substantive synthesis content of this paper.

3.6 Phase-1 closure verdict — honest framing

Honest framing. Phase-1 supplies a **compositional synthesis** of the four QM postulates from the 13 primitives via sixteen Phase-1 theorems. This is **NOT** a closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger sense. Each constituent theorem declares paper-specific postulates whose substrate-derivations are OPEN. The synthesis is at verdict **M2 (Intermediate Path C)** — substrate mechanisms identified across the four postulates; constructive derivations OPEN at the per-theorem level. ED sits in the **substrate-ontology** research-program lineage (Wolfram / 't Hooft / causal-set), not the operational-reconstruction lineage.

Numerical content (specific Hamiltonians, specific observable spectra, $\hbar$ value) is INHERITED.

3.7 What is NOT covered by this synthesis

- **Relativistic generalization:** Phase-1 covers non-relativistic QM. Relativistic content lives in the Arc-R / Wave-3 RQM-bridge.
- **Many-body / QFT:** multi-particle Fock-space content extends Phase-1 (Paper_RQM_Q7Q8 + standard QFT).
- **Open-system / decoherence:** Lindblad (Paper_024) extends to V5-modulated regimes.
- **Specific values of constants ($\hbar$, particle masses):** all VALUE-INHERITED.

4 Audit Table

#	Step	Label	Notes
1	13 primitives	P	Paper_087.
2	Participation measure $P_K = b_K e^{i\pi K}$	I	Paper_U1.
3	Sesquilinear inner-product on motif space	I	Paper_U2.
4	Hilbert-space emergence	I	Paper_007.
5	Born-Gleason rule	I	Paper_002.
6	Stone's theorem applied to substrate translations	I	Papers_U3, U5.
7	Self-adjoint observables	I	Papers_U3, U5.
8	Unitary evolution	I	Papers_006, U3.
9	Projective measurement	I	Paper_005.
10	Postulate I derived	D-via-I	Composition.
11	Postulate II derived	D-via-I	Composition.
12	Postulate III derived	D-via-I	Composition.
13	Postulate IV derived	D-via-I	Composition.
14	Four postulates compositionally synthesized	D-via-I	Not reductive.
15	Numerical content	I	Empirical.
16	Per-theorem closure across 16 Phase-1 theorems	OPEN	Synthesis only.
17	Verdict M2	A→position	Per Paper_095.

5 Falsification Criteria

- **F1:** Any constituent Phase-1 derivation (Papers_002, 005, 006, 007, U1–U5) is refuted at its F-level — propagates upward to refute this synthesis.

- **F2:** Empirical demonstration that the four QM postulates require additional independent structural content beyond the participation measure + kernel calculus — would weaken the unification claim.
- **F3:** Discovery of a substrate-level QM phenomenon not derivable from participation measure + V1/V5 — would force extension.

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

This paper does **not** claim Hardy 2001 / CDP 2011 / Coecke-Kissinger-style closed-proof reconstruction of the four QM postulates. The synthesis is compositional, with per-theorem substrate-derivations OPEN.

This paper is the **synthesis capstone** of the Phase-1 QM-emergence program. It integrates Papers_002, 005, 006, 007, U1–U5 into a single unified-structure account of the four QM postulates.

Numerical content INHERITED. All numerical content (Hamiltonians, observable spectra, $\hbar$). **Form COMPOSED.** The four-postulate unification under participation measure + kernel calculus.

Future work. Substrate-level synthesis of relativistic QM (companion Wave-3 RQM-bridge synthesis paper). Substrate audit of axiomatic alternatives (Hardy, CDP, Coecke-Kissinger reconstructions) for cross-validation.

Verdict: **M2 (compositional synthesis with declared per-theorem postulates).**